

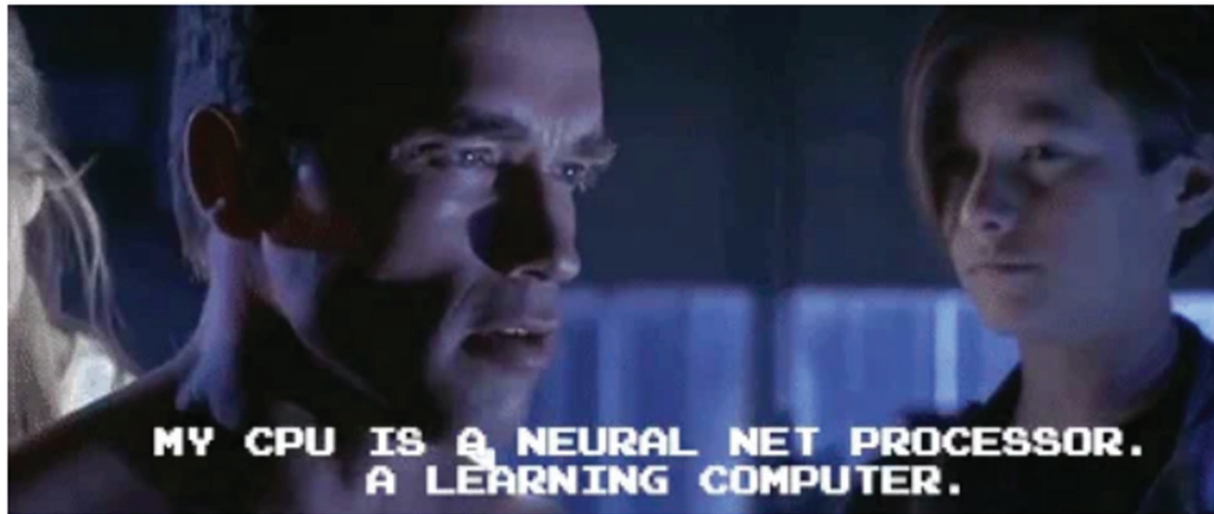
# Fundamentals of Machine Learning (Beginner)

24<sup>th</sup> May – 25<sup>th</sup> May 2023

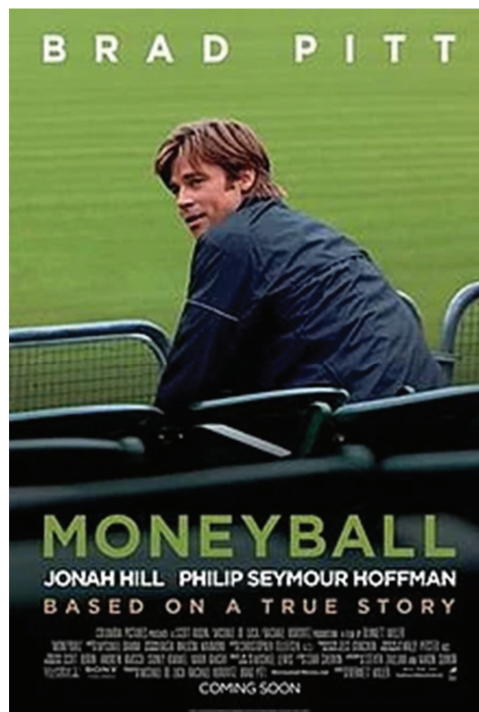
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## A SOFT INTRODUCTION TO MACHINE LEARNING

## Machine Learning: An Overview

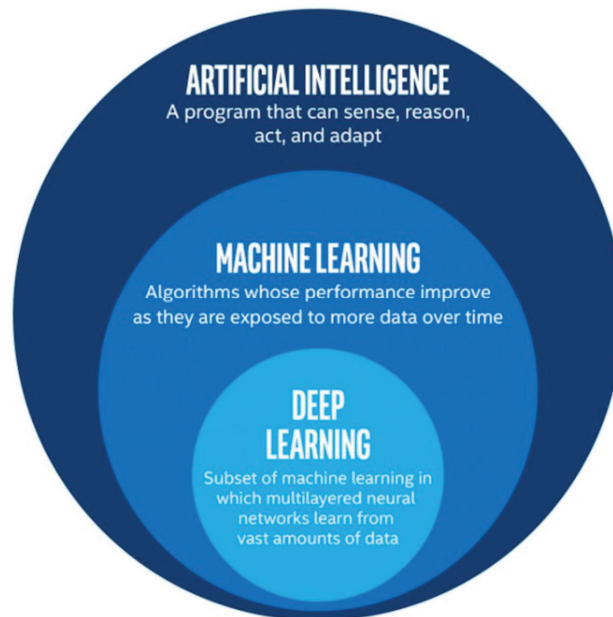


## Machine Learning: An Overview



# Machine Learning: An Overview

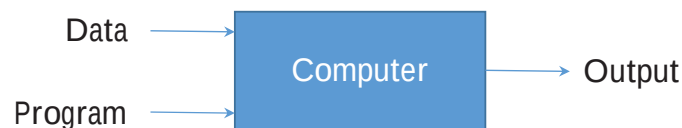
- What is Machine Learning?



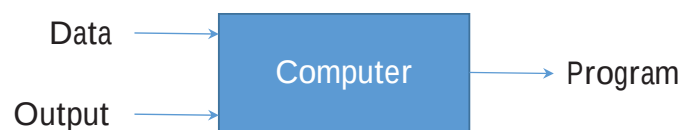
## WHAT IS MACHINE LEARNING?

- Machine learning is the field of study that gives computers the ability to learn without being programmed explicitly.
- Machine learning focuses on the development of computer programs that can access data and use it to learn by themselves.

### Traditional Programming

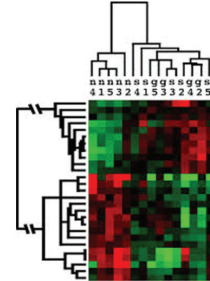


### Machine Learning



# WHEN DO WE USE MACHINE LEARNING?

- Machine Learning is used when:
  - Human expertise does not exist (navigating on Mars)
  - Humans can't explain their expertise (speech recognition)
  - Models must be customized (personalized medicine)
  - Models are based on huge amounts of data (genomics)



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# WHEN DO WE USE MACHINE LEARNING?

- Some more examples of tasks that are best solved by using a learning algorithm
  - Recognizing patterns:**
    - Facial identities or facial expressions
    - Handwritten or spoken words
    - Medical images
  - Generating patterns:**
    - Generating images or motion sequences
  - Recognizing anomalies:**
    - Unusual credit card transactions
    - Unusual patterns of sensor readings in a nuclear power plant
  - Prediction:**
    - Future stock prices or currency exchange rates

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## SAMPLE APPLICATIONS

- Web search
- Computational biology
- Finance
- E-commerce
- Space exploration
- Robotics
- Information extraction
- Social networks
- Debugging software

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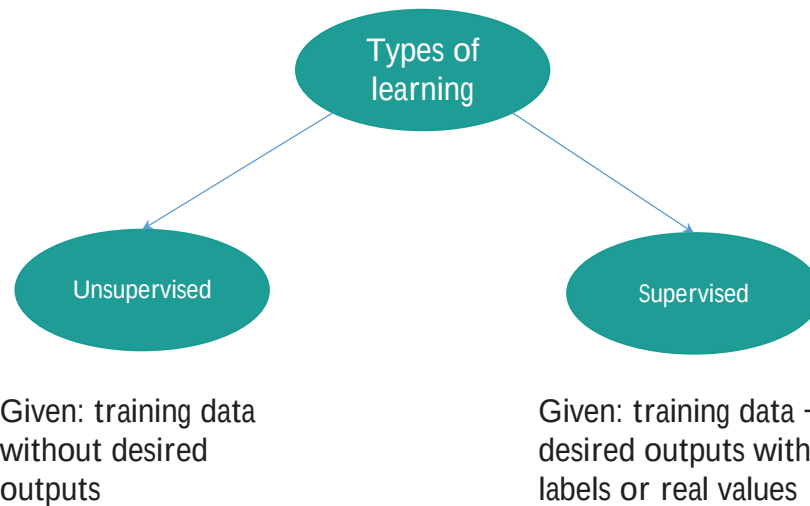
## Check your understanding

- In machine learning, computer learns by itself by looking at the examples given. We do not hard code every possible case.
  - A. True
  - B. False

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# TYPES OF MACHINE LEARNING



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# TYPES OF MACHINE LEARNING

## 1) Supervised learning

- Supervised learning algorithms try to model relationships and dependencies between the target prediction output and the input features such that we can predict the output values for new data based on those relationships which it learned from the previous data sets
- We have a set of features or inputs  $X$  (for example, an image) and our model will predict a target or output variable  $y$ 
$$y=f(X)$$
- Features are independent variables and targets are the dependent variable.
- Can further be grouped into Classification and Regression problems
  - Classification – Spam filtering or image classification
  - Regression - Given some information about a house and predict its price

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# TYPES OF MACHINE LEARNING

## 2) Unsupervised learning

- Mainly used in pattern detection and descriptive modeling. However, there are no output categories or labels here based on which the algorithm can try to model relationships.
- We have input data  $X$  but no output variable  $y$ . Goal of unsupervised learning is to model the distribution of the data, i.e. identify patterns in the data.
- Can be further divided into Clustering and Association problems.
  - **Clustering**
    - Anomaly Detection: Given measurements from sensor in a manufacturing facility
    - Given a list of customer and information about them and discover which group of users do they fall into
  - **Association**
    - Discover patterns in data such as whenever it rains, people tend to stay indoors. When it is hot, people buy more ice-cream

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## Check your understanding

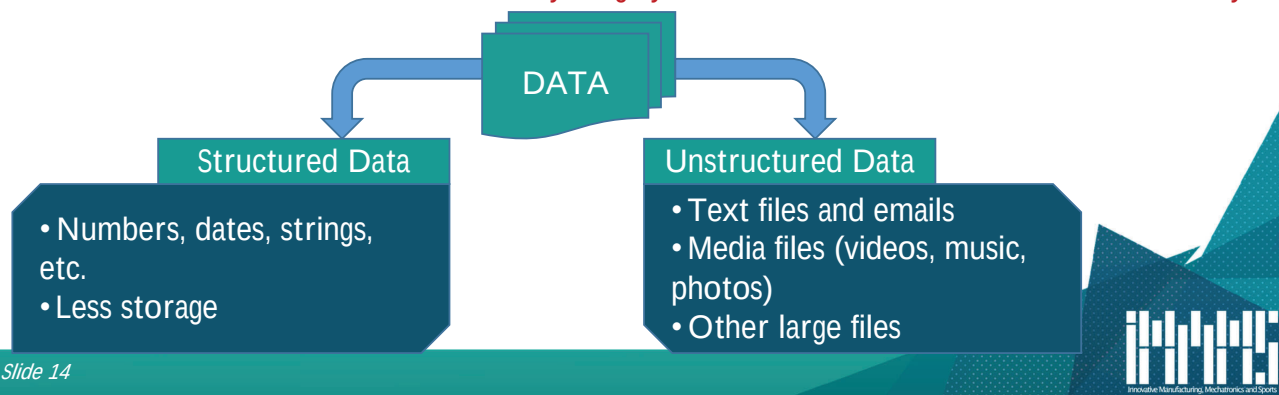
- Given previous prices and news about a company, predicting the company's stock prices at the end of next week.
  - A. Regression
  - B. Classification
  - C. Clustering
- Based on user's activity (e.g. browsing movies Netflix), we want to group users in-order to understand the various market segments.
  - A. Regression
  - B. Classification
  - C. Clustering
- Predicting whether a person is male or female, given height, weight and other parameters.
  - A. Regression
  - B. Classification
  - C. Clustering

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# DATA ACQUISITION

- Machine learning needs two things to work, data (lots of it) and models.
- When acquiring the data, be sure to have enough features to train correctly your learning model. In general, the more data you have the better so make to come with enough rows.
- Data is generally stored as a CSV file (comma separated value file created with excel) or any other supported type of dataset.
- What kind of data should you collect?
  - Data can be categorized into two types: Structured and Unstructured. Structured Data refers to well-defined types of data that are stored in search-friendly databases, while Unstructured Data is “everything” you can collect — but it's not search-friendly.



# DATA VISUALIZATION

- Data visualization is the representation of data or information in a graph, chart, or other visual format. It communicates relationships of the data with images.
- This is important because it allows trends and patterns to be more easily seen. With the rise of big data upon us, we need to be able to interpret increasingly larger batches of data.
- Machine learning makes it easier to conduct analyses such as predictive analysis, which can then serve as helpful visualizations to present.
- Types of data visualization charts:
  - Line chart
  - Bar chart
  - Histogram
  - Scatter plot
  - Heat map
  - Confusion matrix



# DATA PRE-PROCESSING

## ▪ Data Imputation

- Datasets may have missing values, and this can cause problems for many machine learning algorithms.
- It is good practice to identify and replace missing values for each column in your input data prior to modeling your prediction task.
- Several different approaches to imputing missing values are found in the literature:

### 1. Imputation using mean or most frequent value

- Imputing all missing values with the mean for quantitative variables, or the most common value for categorical variables.

|   | col1 | col2 | col3 | col4 | col5 |          |   | col1 | col2 | col3 | col4 | col5 |
|---|------|------|------|------|------|----------|---|------|------|------|------|------|
| 0 | 2    | 5.0  | 3.0  | 6    | NaN  | mean() → | 0 | 2.0  | 5.0  | 3.0  | 6.0  | 7.0  |
| 1 | 9    | NaN  | 9.0  | 0    | 7.0  |          | 1 | 9.0  | 11.0 | 9.0  | 0.0  | 7.0  |
| 2 | 19   | 17.0 | NaN  | 9    | NaN  |          | 2 | 19.0 | 17.0 | 6.0  | 9.0  | 7.0  |

### 2. Imputation using a randomly selected value

- This works by randomly selecting an observed entry in the variable and use it to impute missing values.

### 3. Remove missing values

- This works by completely remove the rows (data points) with missing values.

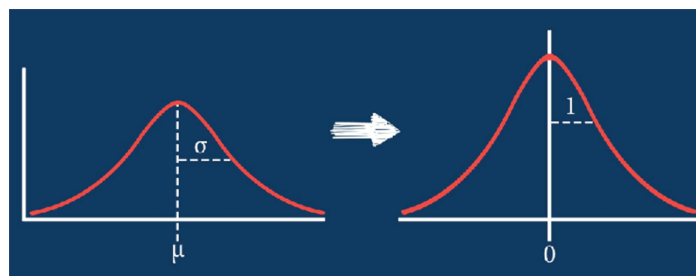
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# DATA PRE-PROCESSING

## ▪ Data Normalization

- Process of efficiently organizing data in a database.
- Normalization is to change the values of numeric columns in the dataset to a common scale, without distorting differences in the ranges of values.
- **Normalizing** a vector most often means dividing by a norm of the vector. It also often refers to rescaling by the minimum and range of the vector, to make all the elements lie between 0 and 1 thus bringing all the values of numeric columns in the dataset to a common scale.
- For machine learning, every dataset does not require normalization. It is required only when features have different ranges.

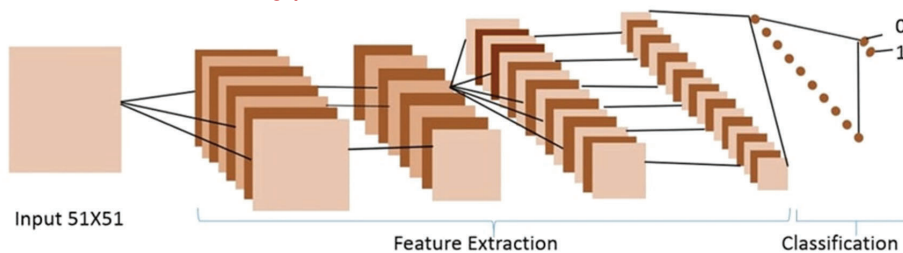


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# FEATURE EXTRACTION

- Feature extraction is a process of dimensionality reduction by which an initial set of raw data is reduced to more manageable groups for processing.
- Feature extraction is the methods that select and /or combine variables into features, effectively reducing the amount of data that must be processed, while still accurately and completely describing the original data set.
- Why is feature extraction useful?
  - The process of feature extraction is useful when you need to reduce the number of resources needed for processing without losing important or relevant information.
  - Feature extraction can also reduce the amount of redundant data for a given analysis. Also, the reduction of the data and the machine's efforts in building variable combinations (features) facilitate the speed of learning and generalization steps in the machine learning process.



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# DATA SAMPLING

- Splitting data
  1. Training dataset
    - The sample of data used to fit the model.
  2. Test dataset
    - The sample of data used to provide an unbiased evaluation of a final model fit on the training dataset.
  3. Validation dataset
    - The sample of data used to provide an unbiased evaluation of a model fit on the training dataset while tuning model hyperparameters. The evaluation becomes more biased as skill on the validation dataset is incorporated into the model configuration.
- Make sure that your test set meets the following two conditions:
  - Is large enough to yield statistically meaningful results.
  - Is representative of the data set as a whole. In other words, don't pick a test set with different characteristics than the training set.



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## Check your understanding

- You are given a dataset of 10,000 images - each image being either of a horse or a donkey. Which of these is the best way to split the data into train and test?
  - A. All horses are in train, and all donkeys in test
  - B. All donkeys are in train, and all horses in test
  - C. Use the entire data for training and also testing
  - D. Randomly choose each data point to be in train or test

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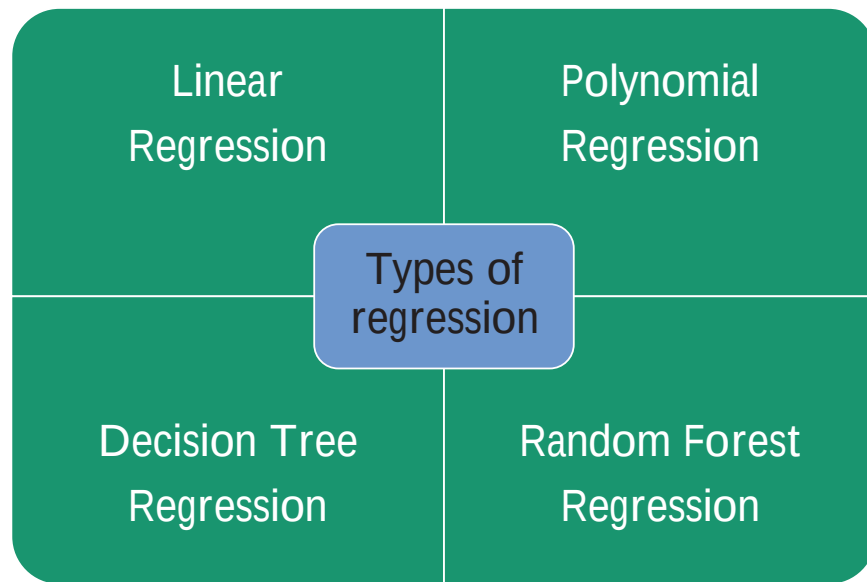
## REGRESSION ANALYSIS

- Regression is the process of finding a model that predicts a continuous value based on its input variables. In regression problems, the goal is to mathematically estimate a mapping function ( $f$ ) from the input variables ( $x$ ) to the output variables ( $y$ ).
- It is a statistical method to model the relationship between a dependent (target) and independent (predictor) variables with one or more independent variables.
- More specifically, regression analysis helps us to understand how the value of the dependent variable is changing corresponding to an independent variable when other independent variables are held fixed.
- A regression algorithm is commonly evaluated by calculating the **root mean squared error (RMSE)** of its output.

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# REGRESSION ANALYSIS



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# REGRESSION ANALYSIS

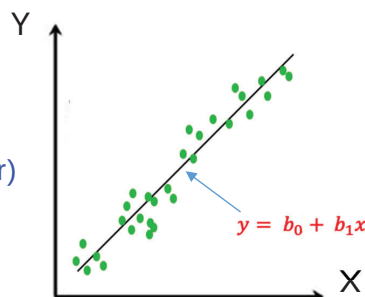
## 1. Linear Regression

- Linear regression is a supervised learning. It's a statistical regression method which is used for predictive analysis.
- Linear regression shows the linear relationship between the independent variable (X-axis) and the dependent variable (Y-axis), hence called linear regression.
- If there is only one input variable (x), then such linear regression is called simple linear regression. And if there is more than one input variable, then such linear regression is called multiple linear regression.
- Linear Regression establish relationship between independent and dependent variables by fitting a best line. This best fit line is known as regression line and represented by a linear equation:

$$y = b_0 + b_1x$$

– In this equation:

- y – Dependent Variable (target)
- x – Independent variable (predictor)
- b – linear coefficients



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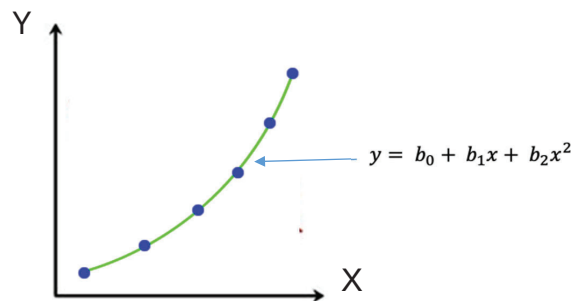
# REGRESSION ANALYSIS

## 2. Polynomial Regression

- A type of regression which models the non-linear dataset using a linear model.
- It is similar to multiple linear regression, but it fits a non-linear curve between the value of x and corresponding conditional values of y.
- In Polynomial regression, the original features are transformed into polynomial features of given degree and then modeled using a linear model. Which means the datapoints are best fitted using a polynomial line.
- The equation for polynomial regression also derived from linear regression equation.

$$y = b_0 + b_1x + b_2x^2 + \dots + b_nx^n$$

- The model is still linear as the coefficients are still linear with quadratic.



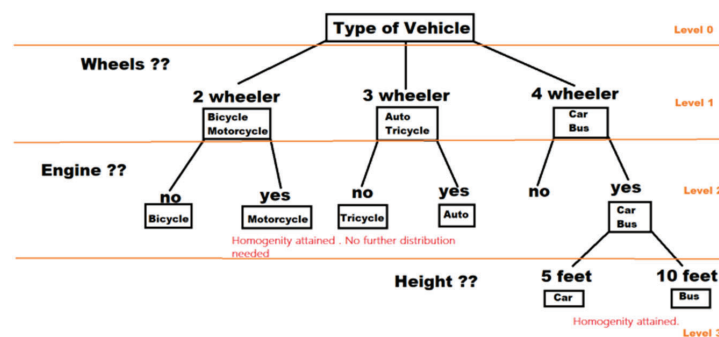
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# REGRESSION ANALYSIS

## 3. Decision Tree Regression

- A supervised learning algorithm which can be used for solving both classification and regression problems.
- It can solve problems for both categorical and numerical data
- Decision Tree regression builds a tree-like structure in which each internal node represents the "test" for an attribute, each branch represent the result of the test, and each leaf node represents the final decision or result.
- A decision tree is constructed starting from the root node/parent node (dataset), which splits into left and right child nodes (subsets of dataset). These child nodes are further divided into their children node, and themselves become the parent node of those nodes.
- For example, the decision is to be made to predict the vehicle name according to the properties like number of wheels , height of vehicle and presence of engine .



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## Check your understanding

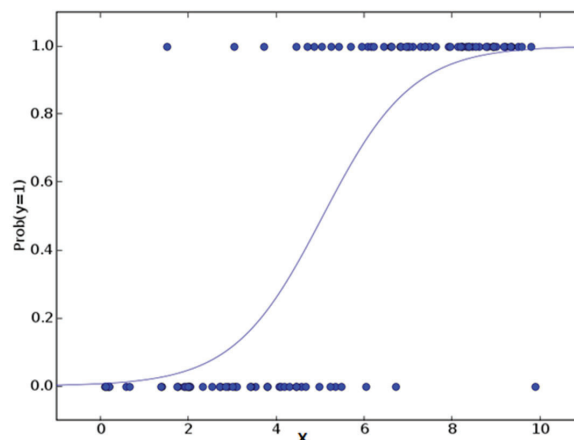
1. Is Linear Regression is a supervised machine learning algorithm?
  - a) True
  - b) False
2. Is Linear Regression is mainly used for Regression?
  - a) False
  - b) True
3. Which of the following evaluation metrics can be used to evaluate a model while modeling a continuous output variable?
  - a) AUC-ROC
  - b) Accuracy
  - c) Logloss
  - d) Root Mean Squared Error

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## MACHINE LEARNING MODELS

1. Logistic Regression
  - It is used to estimate discrete values ( Binary values like 0/1, yes/no, true/false ) based on given set of independent variable(s).
  - It predicts the probability of occurrence of an event by fitting data to a logit function. Hence, it is also known as logit regression. Since, it predicts the probability, its output values lies between 0 and 1 (as expected)



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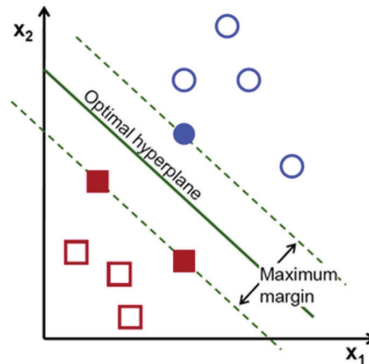




# MACHINE LEARNING MODELS

## 2. Support Vector Machine (SVM)

- Each data item is plotted as a point in n-dimensional space (where n is number of features you have) with the value of each feature being the value of a particular coordinate.
- The objective of the SVM algorithm is to find a hyperplane in n-dimensional space (n is the number of features) that distinctly classifies the data points.
- To separate the two classes of data points, many possible hyperplanes could be chosen. Maximizing the margin distance provides some reinforcement so that future data points can be classified with more accuracy.



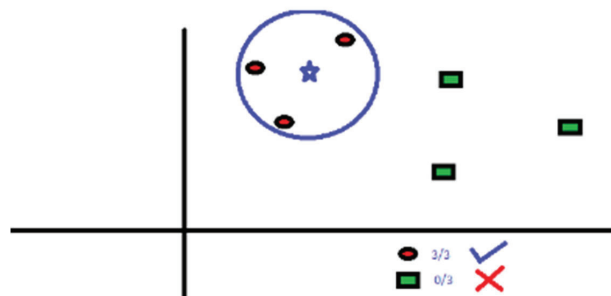
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# MACHINE LEARNING MODELS

## 3. k-Nearest Neighbours (kNN)

- kNN is a simple algorithm that stores all available cases and classifies new cases by a majority vote of its k neighbors. The case being assigned to the class is most common amongst its kNN measured by a distance function.
- These distance functions can be Euclidean, Manhattan, Minkowski and Hamming distance.
- First three functions are used for continuous function and fourth one (Hamming) for categorical variables. If  $K = 1$ , then the case is simply assigned to the class of its nearest neighbor.



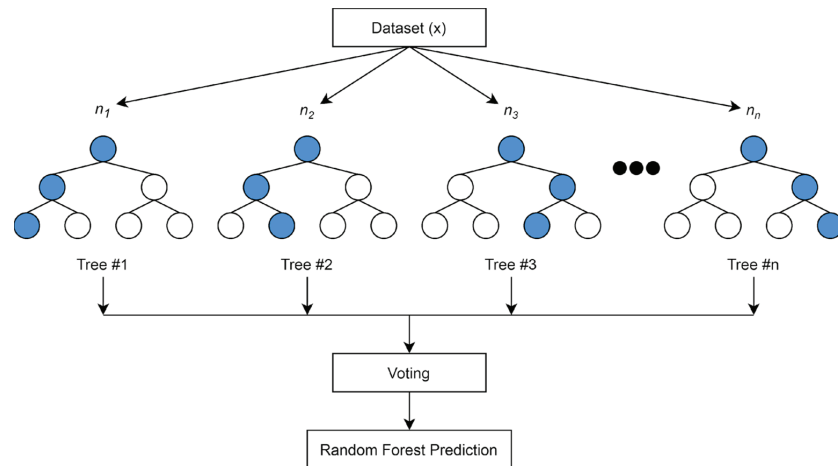
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# MACHINE LEARNING MODELS

## 4. Random Forest

- Random Forest is a trademark term for an ensemble of decision trees. In Random Forest, we have a collection of decision trees (so known as “Forest”).
- To classify a new object based on attributes, each tree gives a classification and we say the tree “votes” for that class. The forest chooses the classification having the most votes (over all the trees in the forest).



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## MODEL PERFORMANCE EVALUATION

### Confusion Matrix

|        |     | Predicted      |                |
|--------|-----|----------------|----------------|
|        |     | Yes            | No             |
| Actual | Yes | True Positive  | False Negative |
|        | No  | False Positive | True Negative  |

### Performance evaluation:

#### 1) Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

#### 2) Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

#### 3) Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

#### 4) F1-Test

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

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# SELECTION OF BEST MODEL

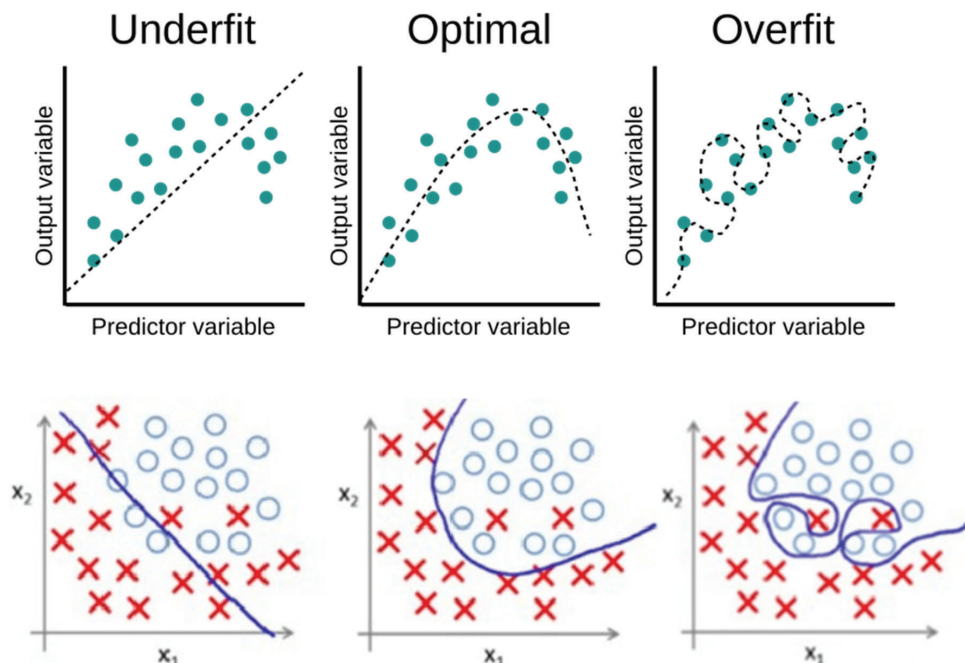
- Generalisation
  - the ability of a machine learning model to perform well on new unseen data
- Overfitting
  - Overfitting refers to a model that models the training data too well.
  - Overfitting happens when a model learns the detail and noise in the training data to the extent that it negatively impacts the performance of the model on new data.
  - This means that the noise or random fluctuations in the training data is picked up and learned as concepts by the model. The problem is that these concepts do not apply to new data and negatively impact the models ability to generalize.
- Underfitting
  - Underfitting refers to a model that can neither model the training data nor generalize to new data.
  - An underfit machine learning model is not a suitable model and will be obvious as it will have poor performance on the training data.

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# SELECTION OF BEST MODEL

- Comparison on underfitting vs. optimal vs. overfitting

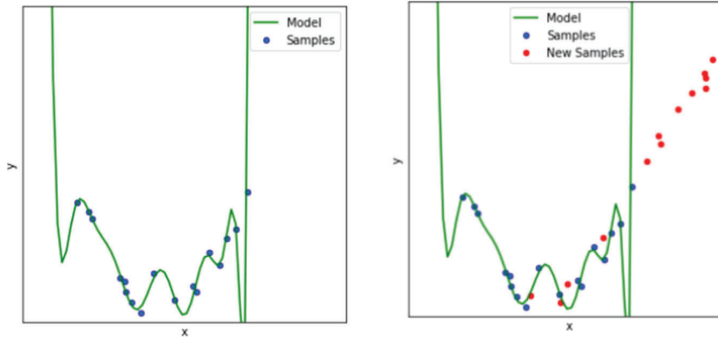


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## Check your understanding

- In the graphs on the left below we can see a polynomial regression model curve (in green) which can predict almost every point (blue circles) on the training set exactly. But when new points (orange circles) are introduced in the graph to the right, it is unable to make accurate predictions on most of them.



- How you would you describe the model?
  - A. Underfitting
  - B. Overfitting
  - C. Neither, it generalizes perfectly well

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## FEATURE SELECTION

- Feature Selection is the process where you automatically or manually select those features which contribute most to your prediction variable or output in which you are interested in.
- Having irrelevant features in your data can decrease the accuracy of the models and make your model learn based on irrelevant features.
- Benefits of performing feature selection before modeling your data:
  - Reduces overfitting
    - Less redundant data means less opportunity to make decisions based on noise.
  - Improves accuracy
    - Less misleading data means modeling accuracy improves.
  - Reduces training time
    - Fewer data points reduce algorithm complexity and algorithms train faster.

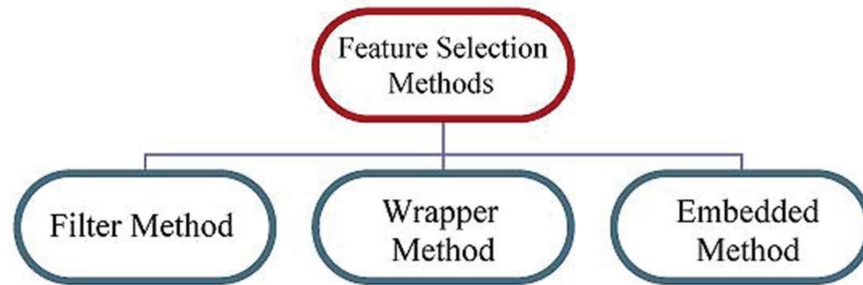
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# FEATURE SELECTION

- Types of feature selection:



## 1. Filter Method

- Filter feature selection methods apply a statistical measure to assign a scoring to each feature. The features are ranked by the score and either selected to be kept or removed from the dataset.
- The methods are often univariate and consider the feature independently, or with regard to the dependent variable.
- Some examples of some filter methods include the information gain, information gain ratio, Gini decrease, ANOVA, Chi-squared test and correlation coefficient scores.

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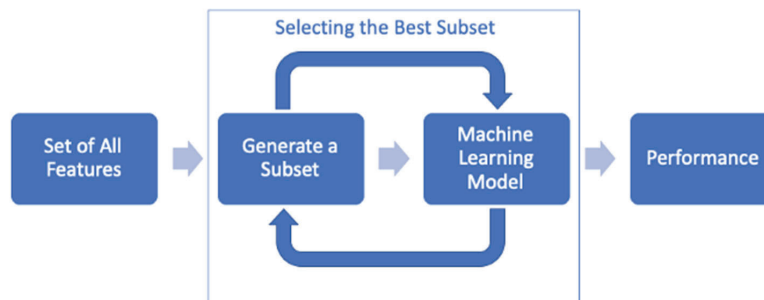
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# FEATURE SELECTION

## 2. Wrapper methods

- Wrapper methods consider the selection of a set of features as a search problem, where different combinations are prepared, evaluated and compared to other combinations.
- A predictive model is used to evaluate a combination of features and assign a score based on model accuracy.



- Examples of wrapper search methods:

- Sequential Forward Selection
- Sequential Backward Selection
- Bi-directional elimination

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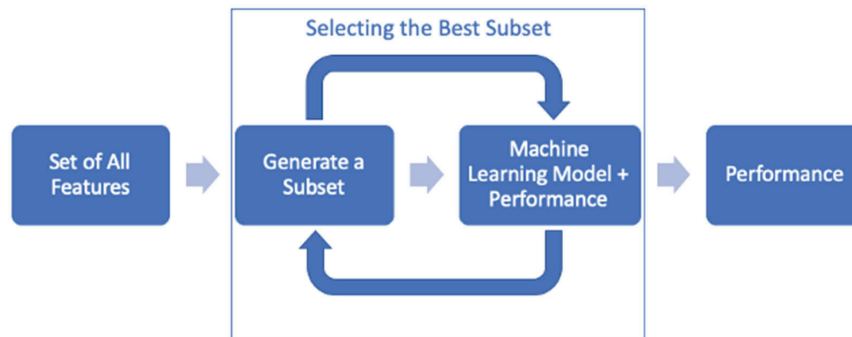
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# FEATURE SELECTION

## 3. Embedded methods

- The Embedded Method includes its own feature selection methods such as Tree-based Feature Importance. In other words, we don't need to select features yourself.
- Measure (Indicator)
  - Classification – Information Gain/Entropy OR Gini Impurity
  - Regression Trees - Variance



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## General Workflow

- Load the dataset.
- Split the dataset into train dataset, and test dataset. Typically, we do a 70:30 ratio. (Depending on the size of the dataset)
- Train the model using the train dataset by using the default hyperparameters of a given model.
- Evaluate the model on the test dataset.
- Based on how much the model is overfitting (IF), decide on the significant features that could provide a better generalization. Retrain the model based on the features.
- Test the final model on the test dataset.

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The slide features a teal background with abstract geometric shapes. A large, light gray circle is centered on the slide, serving as a backdrop for the main text.

# THANK YOU